Discover Sustainability

Review

From automation to collaboration: exploring the impact of industry 5.0 on sustainable manufacturing

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Abstract
Industry 5.0 represents a significant shift from the automation-centric paradigm of Industry 4.0, emphasizing human–machine collaboration, sustainability, and resilience in industrial operations. This study explores the integration of advanced technologies such as artificial intelligence (AI), collaborative robotics, human-digital twins (HDTs), and the Internet of Things (IoT) with human creativity to drive sustainable and adaptive manufacturing processes. Through a systematic literature review (SLR) of 110 articles, this research identifies key drivers, challenges, and opportunities associated with Industry 5.0. The findings highlight Industry 5.0's role in enhancing mass customization, reducing environmental impact through circular economy principles, and improving workforce adaptability through reskilling initiatives. Despite its potential, challenges such as technological integration, ethical AI concerns, regulatory barriers, and cybersecurity risks remain critical hurdles to adoption. This study also proposes a structured Industry 5.0 framework, outlining key implementation steps, including human-technology integration, ethical compliance, sustainable manufacturing strategies, workforce adaptation, and real-time optimization. The research contributes to existing literature by providing a comparative analysis of Industry 4.0 and Industry 5.0, testable hypotheses for future empirical validation, and insights into policy and industrial strategies for seamless transition. The study concludes that Industry 5.0 is not just an extension of Industry 4.0 but a transformative paradigm aimed at fostering a sustainable, human-centric industrial future. The findings provide actionable strategies for industry leaders, policymakers, and researchers to navigate the evolving industrial landscape effectively.

Keywords Industry 5.0 · TBL · Sustainable manufacturing · Human–machine collaboration · Advanced technologies

1 Introduction

The manufacturing industry is undergoing a significant transformation, driven by the rapid adoption of advanced technologies and evolving societal needs. From the mechanized processes of the first industrial revolution to the digitalization that characterizes Industry 4.0 (I4.0), the world has witnessed unprecedented changes in manufacturing practices [1]. Now, as we transition into the next phase, Industry 5.0 (I5.0) promises a paradigm shift that blends technological innovation with human-centric approaches, aiming to address the shortcomings of I4.0 while fostering sustainability across all dimensions [2]. Industry 5.0 is more than a mere extension of previous industrial revolutions—it represents

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a new era in which technology and human creativity collaborate to achieve personalized, efficient, and sustainable manufacturing (SM) solutions [3].

Industry 5.0 integrates cutting-edge technologies such as Artificial Intelligence (AI), the Internet of Things (IoT), collaborative robotics (cobots), and big data analytics [4]. While I4.0 focused on automation and data-driven efficiency, I5.0 emphasizes the importance of human involvement in the production process, mass customization, and a strong commitment to sustainability. A key principle that underpins Industry 5.0 is the concept of Triple-Bottom-Line (TBL) sustainability, which balances economic, social, and environmental concerns to ensure long-term viability [5]. By reintegrating humans into the manufacturing process and promoting ethical, resource-efficient practices, I5.0 presents an opportunity to address not only technological challenges but also global issues such as climate change, resource depletion, and job displacement [6].

The Industrial Revolution began with Industry 1.0 in the late eighteenth century, marked by mechanization using steam power and the development of the first power-driven machines [7]. When Edmund Cartwright invented the first power loom in 1784, it greatly increased the efficiency and output of the weaving process, marking a watershed moment in the mechanization of the textile industry. Industry 2.0, in the early twentieth century, introduced mass production and assembly lines powered by electricity, revolutionizing manufacturing efficiency [8]. Automation in production was made possible in Industry 3.0 by means of electrical and digital technology [9, 10]. The idea behind Industry 4.0 is to improve production efficiency by integrating digital technology into operations in the factory for real-time monitoring and crucial information provision to decision-makers [11, 12]. As a result of this shift, cutting-edge technologies like the IIoT, cyber-physical systems, AI, digital twins, industrial robots, and additive manufacturing have enabled previously unseen levels of automation, connectivity, and productivity [13, 14]. As a direct outcome of integrating several cutting-edge technologies, the concept of "smart factories" is central to Industry 4.0 [15, 16].

This change has encountered several hurdles. Rapid advancements in Industry 4.0 technology have been made, although it has simultaneously introduced other unexpected social and environmental issues, particularly in populous nations such as India, Bangladesh, and Pakistan [17, 18]. Concerns about potential job displacement among workers have been heightened by the deployment of I4.0 technologies, according to a research [17]. In addition, their findings show that low-skilled workers are hit particularly hard, especially in the manufacturing sector of India, which has used a lot of people with low levels of education and training for a variety of tasks. The necessity to achieve Triple-Bottom-Line (TBL) sustainability has been magnified due to the numerous global challenges brought about by modern industrialization [19].

A comprehensive literature review reveals that several systematic literature reviews (SLRs) [6, 20, 21] establish a robust foundation for elucidating the essential distinctions and progressions between these two industrial revolutions. Several studies highlight the transition from automation and digitalisation in Industry 4.0 to human–machine cooperation in Industry 5.0 [22, 23]. Additional systematic literature reviews underscored the emphasis on sustainability, human centricity, and resilience in Industry 5.0 [17, 24]. The distinguishing feature of this SLR study is the thorough analysis it provides of I5.0 from a TBL sustainability perspective. It highlights the unique qualities and forces behind I5.0, acknowledges the possibilities and challenges, and provides a holistic framework to help with the transition to I5.0.

1.1 Study motivation and research gaps

I5.0 is a crucial opportunity to actively contribute to the industrial industry and promote sustainable growth. When it comes to intelligent and collaborative manufacturing systems that prioritize TBL sustainability, with the release of I5.0, Bangladesh has a chance to become dominant, which would help the country achieve its goal of becoming a global manufacturing hub. Developed countries used to be the exclusive source for cutting-edge technology in the Indian subcontinent. While several researchers have studied I5.0, the impact on TBL sustainability has not been extensively studied [19]. Further research is needed to explore the complicated relationship between human–machine collaboration, technical advancements, and sustainability [25].

This paper analyses the potential of I5.0, including implications, possibilities, and difficulties, to enhance the resilience and sustainability of the Indian manufacturing industry. The purpose of this study is to provide an unbiased assessment of the I5.0 ecosystems and to provide insights on the subsequent research questions.

Research Question 1: How can we define I5.0 and what elements make it suitable for SM implementation?

Research Question 2: Which technologies make it possible for I5.0 to happen, and how do these technologies affect the long-term viability of TBL?

Research Question 3: across different industries, what possibilities and challenges does Industry 5.0 present?

Research Question 4: How can an all-encompassing sustainable manufacturing (SM) framework be developed by leveraging Industry 5.0 principles and technologies?

The residual manuscript is organized into the subsequent sections: Sect. 2 explores the literature research approach and presents an exhaustive literature evaluation within the subject topic. Section 3 discusses the I5.0's challenges and opportunities. Section 4 explores the impact of I5.0 on sustainable manufacturing. Rest of the sections addresses the implications, limits of this work, and delineates the potential for future research and conclusion, accordingly.

2 Literature review

The foundation of each rigorous and high-quality research project is a thorough literature evaluation of relevant works [26]. So, to answer the research questions stated in the introduction, the literature relevant to the topic was sought for, evaluated, and synthesized using the SLR method.

2.1 Methodology of literature review

Here we see how the SLR technique was used to conduct an in-depth review to understand how I5.0 promotes sustainability in the industrial sector. A SLR guarantees the organized gathering of relevant research data about a particular topic, ensuring consistency with established research questions (RQs) [27]. This systematic literature review technique generates pertinent both quantitative and qualitative conclusions by pulling data from many research papers obtained from several databases [28]. Systematic literature reviews exhibit a significant level of specificity, unlike the broader scope of conventional literature review methodologies, and include specific elements designed to meet the particular research questions of the study, highlighting transparency and inclusion [29].

2.2 Systematic literature review (SLR) methodology

To ensure a structured and repeatable approach in analyzing Industry 5.0 literature, we conducted a Systematic Literature Review (SLR) following established guidelines. The SLR was designed to capture the latest advancements, emerging trends, and challenges in Industry 5.0 by selecting high-quality peer-reviewed studies from recognized databases.

2.2.1 Search strategy and databases

The literature search was conducted across multiple reputable databases, including Scopus, Web of Science, IEEE Xplore, ScienceDirect, and SpringerLink. The search strings were carefully designed to cover a wide range of Industry 5.0-related topics, ensuring a comprehensive dataset. The primary search string used was:

("Industry 5.0" OR "Fifth Industrial Revolution") AND ("collaborative robotics" OR "human digital twin" OR "sustainable manufacturing" OR "ethical AI" OR "human-centric automation").

Additional refinements were made using Boolean operators and database-specific filters, including peer-reviewed journal articles, conference proceedings, and review papers published in the last decade to capture the most relevant research.

2.2.2 Justification for SLR methodology

Several established SLR methodologies were reviewed, including Kitchenham's Guidelines [30] and the PRISMA framework [28]. While these frameworks provide robust methodologies, our study required a tailored approach due to the multidisciplinary nature of Industry 5.0, which spans manufacturing, AI ethics, sustainability, and workforce transformation. Therefore, we adapted key elements from structured SLR methodologies, ensuring comprehensiveness while maintaining flexibility to incorporate emerging trends.

2.2.3 Review questions and literature selection

The SLR was guided by the following key research questions:

What are the core technological advancements driving Industry 5.0?

How does Industry 5.0 address sustainability and human-centric challenges?

What are the key limitations, ethical concerns, and future directions of Industry 5.0 adoption?

The literature selection followed a multi-step process, as outlined in Table 1, including identification, screening, eligibility assessment, and final inclusion based on relevance to the research questions. Each step of this process is explicitly linked to ensure transparency and reproducibility in the review methodology. Using Scopus and the Web of Science databases, previous SLR studies have consistently conducted extensive research within the framework of I4.0 or I5.0 [11, 23].

The research papers retrieved are of higher dependability and quality because of the rigorous indexing methods of journals that are indexed by Web of Science and Scopus. Afterwards, the number was reduced to 55 following initial screening and the application of exclusion and inclusion criteria. The original scope of the search was limited to scholarly journal articles; conference proceedings, book chapters, and editorial comments were not considered. I4.0 came out just a decade ago, and the idea of I5.0 didn't surface until 2017. With a main emphasis on articles published between 2010 and December 2024, this research targets to gather the latest opinions on I4.0, I5.0, SM, and TBL. This is because these industrial paradigms are very new. A small number of research articles published prior to 2010 have been included, with an emphasis on those that are highly pertinent to sustainability and social media, in recognition of the importance of prior research investigations. In the fifth phase, we looked at the references that were included in the articles that were suitable for the study. We then conducted a forward search to locate further publications that were cited in the earlier papers. The increased pool contained 110 relevant publications after this process. We categorized the results according to theoretical perspectives and empirical findings to make sure that these identified publications were properly reported and cited throughout the study.

2.3 Overview of industry 5.0

Industry 5.0 represents a new phase in industrial evolution that shifts the focus from automation and efficiency (as seen in Industry 4.0) to human-centric collaboration, sustainability, and resilience in production systems [31]. This next-generation industrial paradigm emphasizes the integration of advanced technologies with human capabilities to create more personalized, adaptive, and environmentally sustainable manufacturing processes [32, 33]. Unlike the heavily automated, data-driven, and AI-led systems of Industry 4.0, Industry 5.0 brings back the human touch in decision-making processes, re-establishing the value of human expertise, creativity, and customization in industries traditionally dominated by automation [34, 35].

Table 1 The procedures of literature search methodology

Step	Description
Step-1: Articulate Research Objective (RO) and Research Questions (RQs)	Define the main objective of the research (e.g., understanding the role of Industry 5.0 in driving sustainability) and specify the research questions (e.g., RQ1, RQ2, RQ3)
Step-2: Decide Literature Search Strings	Identify keywords for the search: "Industry 5.0" OR "Fifth Industrial Revolution") AND ("collaborative robotics" OR "human digital twin" OR "sustainable manufacturing" OR "ethical AI" OR "human-centric automation"
Step-3a: Apply Criteria for Inclusion (CIs)	Select articles that: 1) Prioritize the incorporation of I5.0 ideas or tools, 2) address sustainability, and 3) are empirical studies. Result: 108 *articles* identified
Step-3b: Apply Criteria for Exclusion (CEs)	Exclude articles that: 1) are conference papers, book chapters, or editorial notes (*14 removed*), 2) were published before 2008 (*16 removed*), or 3) are not written in English (4 *removed*)
Step-4: Consideration of CEs in Research Articles Exclusion	Remove articles based on exclusion criteria, resulting in a refined pool of eligible documents. Total documents excluded: (14 + 16 + 4) = 34. Remaining eligible documents: (108–34) = *74*
Step-5: Backward and Forward Review	Find more relevant articles by reviewing the references in qualifying papers and doing forward searches. Additional articles added: *36* (CI-1: 14, CI-2: 12, CI-3: 10). Later added: *36* Total pools extended to (74 + 36) = *110 articles*
Step-6: Data Extraction and Content Analysis	Categorize the selected and cited articles into theoretical perspectives (*48 articles*) and empirical findings (*62 articles*) for deeper analysis

2.3.1 Key features of industry 5.0

2.3.1.1 Human-centric collaboration The primary feature of Industry 5.0 is the symbiotic relationship between humans and machines. Advanced robotics, AI, and automation technologies work alongside skilled workers, allowing humans to focus on tasks that require creativity, decision-making, and problem-solving [36] as shown in Fig. 1. Automation handles repetitive or hazardous tasks, enabling workers to engage in higher-value roles, fostering job satisfaction, and enhancing overall productivity [37, 38]. This collaboration is facilitated by cobots (collaborative robots), which are designed to work safely and efficiently alongside humans.

2.3.1.2 Customization and personalization Industry 5.0 enables the mass customization of products, a significant shift from traditional mass production. Technologies like 3D printing, AI, and machine learning allow manufacturers to adapt production lines to produce personalized goods at scale [39, 40]. These technologies enable rapid prototyping, shorter product cycles, and the flexibility to tailor products to individual customer needs without sacrificing cost-effectiveness or efficiency. The integration of AI-driven design systems further enhances this ability by optimizing product designs based on consumer preferences and real-time data [41].

2.3.1.3 Sustainability and circular economy A defining feature of Industry 5.0 is its emphasis on sustainability and the circular economy. By integrating green technologies, renewable energy sources, and energy-efficient production systems, Industry 5.0 reduces environmental impact and fosters eco-friendly manufacturing practices [42]. Technologies like IoT enable more efficient use of resources by providing real-time monitoring and predictive maintenance to prevent waste and extend the lifecycle of machinery and products [13, 43]. Furthermore, Industry 5.0 promotes the recycling and repurposing of materials to create a closed-loop production system, which reduces waste and resource consumption [11, 44].

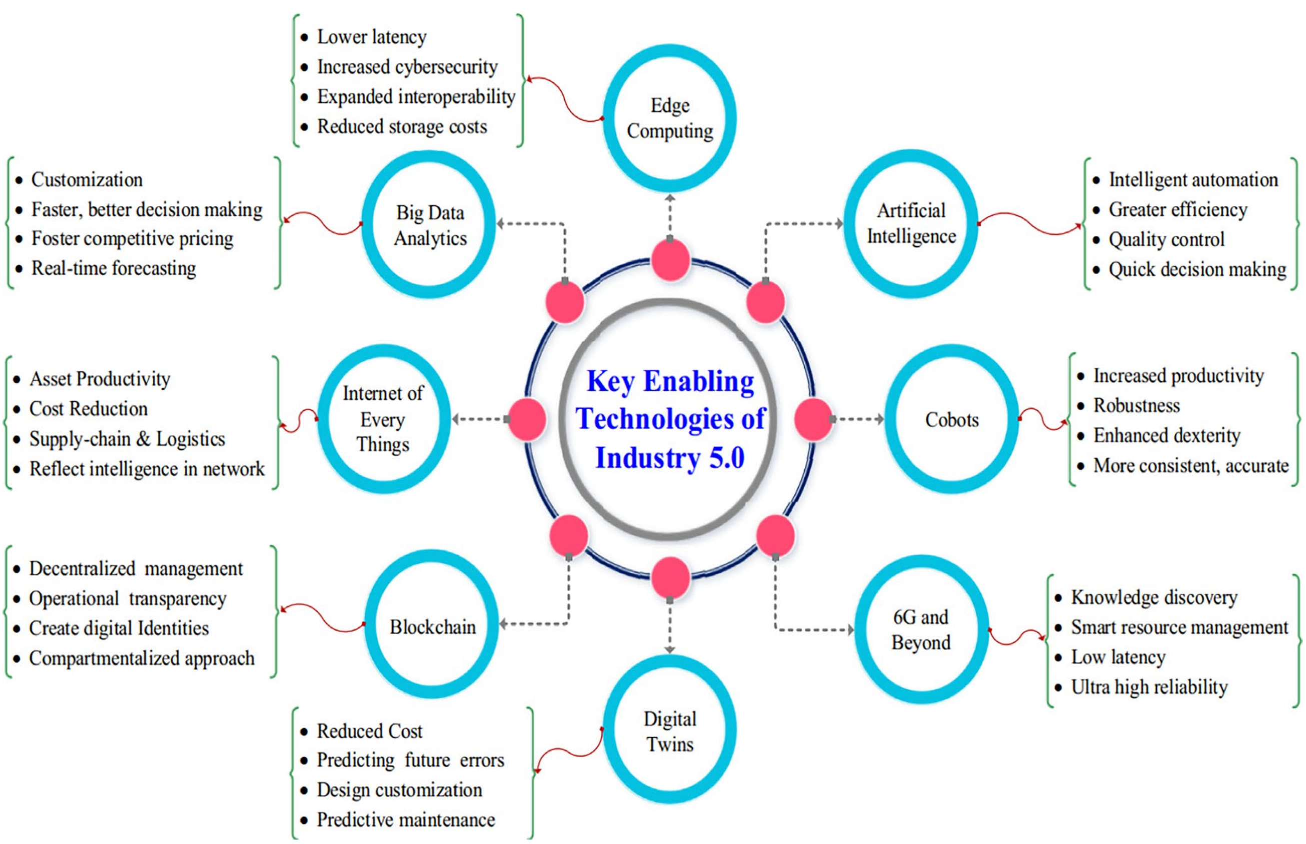

Fig. 1 Key technologies of Industry 5.0 [4]

2.3.1.4 Resilience and adaptability The growing complexity and unpredictability of global supply chains have highlighted the need for adaptable and resilient manufacturing systems. Industry 5.0 enhances these characteristics by incorporating AI, IoT, and blockchain to improve supply chain transparency, track inventory, and enable dynamic adjustments to production schedules [45]. This adaptability allows manufacturers to respond quickly to disruptions, such as supply chain delays or shifts in demand, without compromising quality or operational efficiency [46]. AI-driven predictive analytics and smart factories equipped with IoT sensors help optimize production flow, minimizing downtime and increasing resilience to external shocks [47].

2.3.1.5 Integration of advanced technologies Industry 5.0 builds upon the technological foundation of Industry 4.0 by integrating quantum computing, blockchain, AI, and IoT to improve decision-making, enhance supply chain traceability, and ensure operational transparency [48]. For instance, blockchain ensures secure and transparent data exchange across supply chains, while AI helps with predictive maintenance, quality control, and process optimization [49]. The use of quantum computing in certain aspects of industrial operations will further revolutionize production by enabling faster data processing and solving complex optimization problems that are beyond the reach of traditional computing methods [50].

Industry 5.0 advances human-centric, sustainable, and intelligent manufacturing by integrating collaborative robotics (Cobots), human digital twins (HDTs), and collaborative intelligence. These technologies enhance flexibility, resilience, and sustainability by enabling seamless human–machine collaboration, adaptive decision-making, and improved worker well-being.

Collaborative robotics (Cobots): Cobots work alongside humans, improving flexibility, safety, and productivity in manufacturing. Unlike traditional robots, cobots integrate AI, real-time sensing, and haptic feedback, augmenting human dexterity and reducing ergonomic strain [51]. Cobots support customization, adaptive task allocation, and human–robot collaboration, addressing Industry 4.0's automation limitations [52].

Human digital twins (HDTs): HDTs create real-time digital representations of human workers, optimizing safety, cognitive load, and training [53]. Integrated with factory digital twins, HDTs enable adaptive work environments where systems respond to worker conditions, enhancing efficiency and well-being [54]. However, challenges remain in data privacy and standardization for HDT applications.

Collaborative intelligence (Human-AI Synergy): Collaborative intelligence combines human expertise with AI-driven decision-making, ensuring context-aware, adaptive, and responsible automation [55]. Unlike standalone AI, human-AI teams improve efficiency, quality, and safety in predictive maintenance, smart manufacturing, and mass customization. This approach mitigates AI bias and ethical concerns, ensuring greater transparency in Industry 5.0 [(Collaborative Intelligence in Industry 5.0)].

The integration of cobots, HDTs, and collaborative intelligence fosters resilient, personalized, and intelligent manufacturing. These technologies redefine industrial ecosystems by enhancing human capabilities rather than replacing them, ensuring a sustainable, efficient, and adaptive production model [56].

2.4 The need for industry 5.0

Over the last decade, the global manufacturing sector has made considerable progress toward adopting smart technologies under the framework of Industry 4.0 (I4.0) [38]. This paradigm leverages advancements such as the Internet of Things (IoT), big data analytics (BDA), artificial intelligence (AI), robotics, and cyber-physical systems (CPS) to revolutionize manufacturing processes [57]. The focus has been on automating operations, enhancing efficiency, and enabling real-time communication between interconnected systems. While I4.0 has introduced transformative efficiencies, it has also raised significant concerns regarding social, environmental, and economic sustainability [58]. Challenges associated with I4.0 include widespread apprehensions about job security, increased unemployment due to automation, and environmental issues such as energy overuse, electronic waste, and resource depletion. These concerns highlight the limitations of I4.0, which primarily focuses on economic and technological optimization while overlooking critical social and environmental dimensions [59].

Industry 5.0 (I5.0) emerges as a response to these limitations, offering a more inclusive and balanced approach to industrial development. It highlights the importance of people and robots working together, using cutting-edge tech to boost human ingenuity and creativity [60]. By assigning repetitive or hazardous tasks to robots, I5.0 aims to improve workplace safety while enabling humans to focus on creative and strategic roles. Moreover, it seeks to integrate renewable energy sources and promote the efficient use of resources, aligning industrial practices with global sustainability goals

[61]. Unlike its predecessor, I5.0 prioritizes human-centric values, resilience, and sustainability. It envisions a future where manufacturing processes are not only productive and efficient but also socially and environmentally responsible. By redefining the role of technology to address societal needs and ecological challenges, Industry 5.0 lays the foundation for a more sustainable and equitable industrial era on a global scale [62, 63].

The VOSviewer figure highlights the primary research focus areas within Industry 4.0 (I4.0) as shown in Fig. 2, emphasizing technology-centric themes such as IoT, automation, and digitalization. Larger nodes like "Industry 4.0" and "sustainability" indicate these are significant topics, while smaller nodes for "human-centric approaches" and "circular economy" reveal gaps in social and environmental dimensions. Clusters in the network show interconnected themes, with strong links among technological terms but weaker connections for sustainability-related concepts. This indicates that I4.0 research prioritizes technological advancements over social or ecological aspects. The analysis underscores the need for Industry 5.0 (I5.0) to address these gaps by integrating sustainability, human-centric values, and ecological responsibility into industrial practices.

2.4.1 Some useful case studies

2.4.1.1 Industry-based case study Manufacturing & Production: Margherita & Braccini examined four Italian production organizations that successfully implemented I4.0 technologies. Their findings revealed that while I4.0 improved productivity, it also reduced workforce involvement, raising concerns about job displacement and sustainability. The study emphasized the need for an employee-centric approach to ensure that I4.0 adoption also enhances social sustainability and worker well-being [64]. Strandhagen et al. analyzed four Norwegian businesses and found that the type of produc-

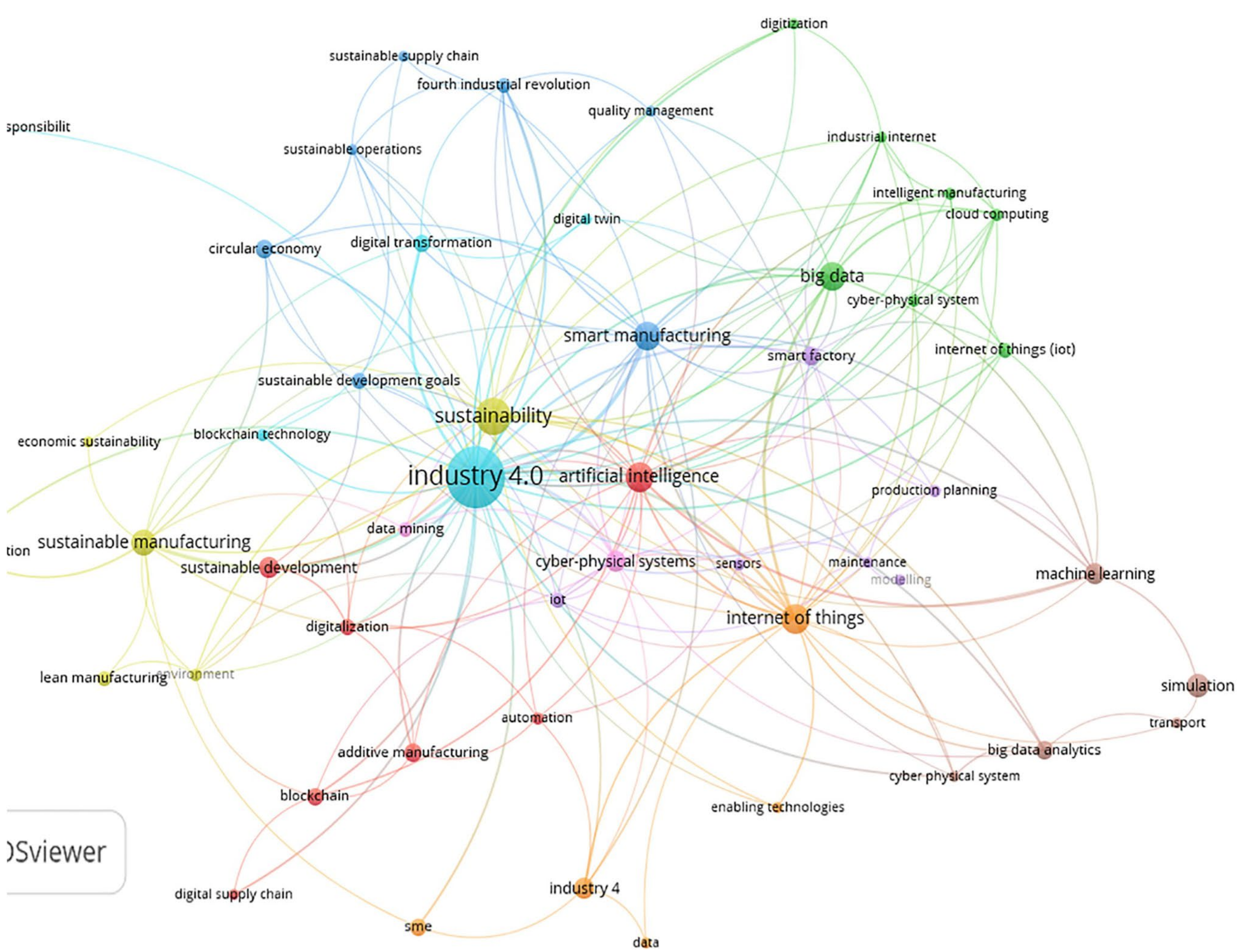

Fig. 2 Most commonly investigated keywords related to Industry 4.0 (Source: Author's creation)

tion system significantly affects I4.0 adoption. Routine manufacturing firms benefit more from automation, whereas batch or work-order-based systems face challenges in implementing I4.0 technologies [65].

Workforce & Employment: Pasi et al. conducted a study on 16 Indian manufacturing firms, revealing that job security concerns remain a major obstacle in I4.0 adoption. Employees feared job losses due to automation, highlighting the need for Industry 5.0's human-centric approach, which seeks to integrate advanced automation while preserving employment opportunities [17].

2.4.1.2 Technology-based Collaborative Robotics & Human–Machine Collaboration: Industry 5.0 shifts the role of robotics from replacing workers to augmenting human capabilities. Cobots will handle repetitive and hazardous tasks, while humans focus on critical thinking and oversight [66]. This transition enhances efficiency, product quality, and worker safety while ensuring greater responsibility and monitoring in automated processes [4].

Sustainability & Renewable Energy: Industry 4.0's environmental impact includes concerns about air pollution, resource depletion, and increased energy consumption [65]. I5.0 addresses these challenges by promoting renewable energy integration and bioenergy adoption in industrial applications [68]. Industry 5.0 redefines industrial priorities by integrating sustainability, societal well-being, and economic development rather than focusing solely on profit-driven objectives. It aims to balance technological progress with environmental responsibility and planetary limits [34, 69].

2.4.1.3 Ethical & regulatory considerations Industry 5.0 emphasizes sustainable industrial ecosystems, ensuring social responsibility, ethical AI, and regulatory compliance in automated decision-making. It prioritizes mass customization, human-AI collaboration, and ethical considerations in smart manufacturing [36, 68].

These case studies highlight the limitations of I4.0 in workforce displacement, sustainability, and production adaptability, while demonstrating how I5.0 addresses these challenges. By integrating human-centric principles, collaborative intelligence, and sustainability-focused technologies, Industry 5.0 creates a resilient and responsible industrial future.

2.5 Comparative analysis: industry 4.0 and industry 5.0

Industry 4.0 (I4.0) has been largely driven by technological and economic objectives, focusing on integrating cutting-edge technologies such as IoT, robotics, AI, and big data into manufacturing systems to increase productivity and flexibility [69]. However, this approach has faced criticism for neglecting critical aspects of sustainability, particularly its environmental and social dimensions [39]. The primary focus on digitalization and automation has often overlooked the human-centric and ecological impacts of industrial operations [34].

In contrast, Industry 5.0 (I5.0) represents a paradigm shift toward value-driven industrial practices, emphasizing the integration of human creativity, sustainability, and resilience. Unlike I4.0, which prioritizes automation and mass production, I5.0 introduces human–robot collaboration and bioeconomy as central themes to address the limitations of its predecessor [37]. It aims to reintroduce humans into the production process, aligning industrial activities with broader sustainability objectives and promoting social and environmental well-being [6].

Several studies have explored the transition from Industry 4.0 to Industry 5.0, yet gaps remain in sustainability integration and practical implementation. Existing research, such as Xu et al. (2021) provides a conceptual discussion on the coexistence of both paradigms but lacks empirical analysis and sustainability considerations [6]. Paschek et al. (2022) examines the transition from Industry 4.0 to Industry 5.0 in the context of Society 5.0 but does not provide industry-specific applications or policy recommendations [70]. Focusing on digital Transition from Industry 4.0 to Industry 5.0 in Smart Manufacturing, Sarkar et al. (2024) analyzed on critical success factors (CSFs) for digital transformation but primarily within the steel manufacturing sector, omitting broader sustainability and workforce implications. In contrast, this study fills these gaps by conducting a systematic literature review (SLR) of 110 articles, offering a structured sustainability framework (Triple Bottom Line—TBL), industry-specific opportunities and challenges, and a collaborative ecosystem model. It uniquely integrates policy recommendations, workforce reskilling strategies, and circular economy principles, ensuring a holistic, human-centric, and sustainability-driven approach to Industry 5.0 adoption. This work provides actionable insights beyond theoretical discussions, addressing economic, social, and environmental sustainability while guiding industries and policymakers toward a smooth Industry 5.0 transition.

This comparison underscores the fundamental differences between the two revolutions, with I4.0 characterized as a techno-economic model and I5.0 as a holistic [71], value-driven approach as shown in Table 2. By addressing the gaps in I4.0, Industry 5.0 paves the way for a more inclusive, sustainable, and human-centric industrial future [41, 69].

Table 2 Comparison between Industry 4.0 and Industry 5.0

Aspect	Industry 4.0	Industry 5.0
Focus	Techno-economic optimization [13]	Value-driven and holistic [67]
Primary Drivers	Automation and efficiency [64]	Human–robot collaboration and sustainability [3]
Key Technologies	IoT, AI, Big Data, CPS, Robotics, 3D Printing [12]	Human–robot collaboration, Bioenergy, IoT, AI, Advanced Materials [68]
Human Involvement	Minimal (reduced human roles)	High (emphasis on human creativity)
Sustainability Goals	Limited (focus on economic gains) [17]	Core objective (Triple-Bottom-Line: Economic, Social, Environmental) [72]
Energy Sources	Mainly electrical energy	Renewable energy and bioenergy [34]
Customization Capability	Mass production [62]	Mass customization [66]
Societal Impact	Concerns over job security, environmental exploitation [62]	Improved work safety, ecological responsibility, social equity [73]

3 Challenges and opportunities of industry 5.0

3.1 Challenges

Industry 5.0 represents a significant shift in industrial practices, blending advanced technologies with human-centric values. However, this transition comes with its own set of challenges as shown in Fig. 3. From ensuring seamless technological integration to addressing ethical concerns and regulatory barriers, organizations must overcome various hurdles to fully realize the potential of Industry 5.0. These challenges are interconnected and require strategic planning, workforce readiness, and investment in innovation to ensure successful implementation.

Technological integration: Combining cutting-edge technologies such as AI, IoT, robotics, and human labor into a cohesive system is a complex task. It requires a significant upfront investment in hardware, software, and infrastructure to ensure seamless interaction between humans and machines [74]. Achieving real-time data exchange, intelligent decision-making, and high efficiency demands the integration of multiple systems, which can be technologically and logistically daunting [75].

Workforce training and adaptation: Transitioning to I5.0 necessitates a workforce skilled in emerging technologies. Employees must learn to program and operate collaborative robots (cobots), manage IoT networks, and interpret large datasets produced by advanced analytics tools. This requires substantial investment in training programs and may lead to resistance among workers who are unprepared or unwilling to adapt to new roles and responsibilities [76, 77].

Ethical concerns and social implications: While I5.0 promotes human-centric approaches, concerns about job displacement persist as automation takes over repetitive tasks. Striking a balance between leveraging AI-driven systems and ensuring equitable employment opportunities poses a significant challenge. Additionally, ethical dilemmas arise in areas such as data privacy, AI bias, and decision-making accountability, further complicating the implementation [78, 79].

Cybersecurity risks: The increased connectivity and reliance on digital technologies expose manufacturing systems to potential cyber threats. Industrial systems equipped with IoT, CPS, and AI are vulnerable to attacks that could disrupt operations, steal sensitive data, or compromise safety. Developing robust cybersecurity measures is critical to ensuring the integrity and reliability of interconnected systems [80, 81].

Regulatory and policy barriers: The rapid pace of technological advancements often outstrips the development of regulatory frameworks [82]. Countries and industries must navigate varying global standards and policies related to data security, ethical AI usage, and environmental regulations. This lack of harmonization can slow down adoption and complicate cross-border collaborations [83].

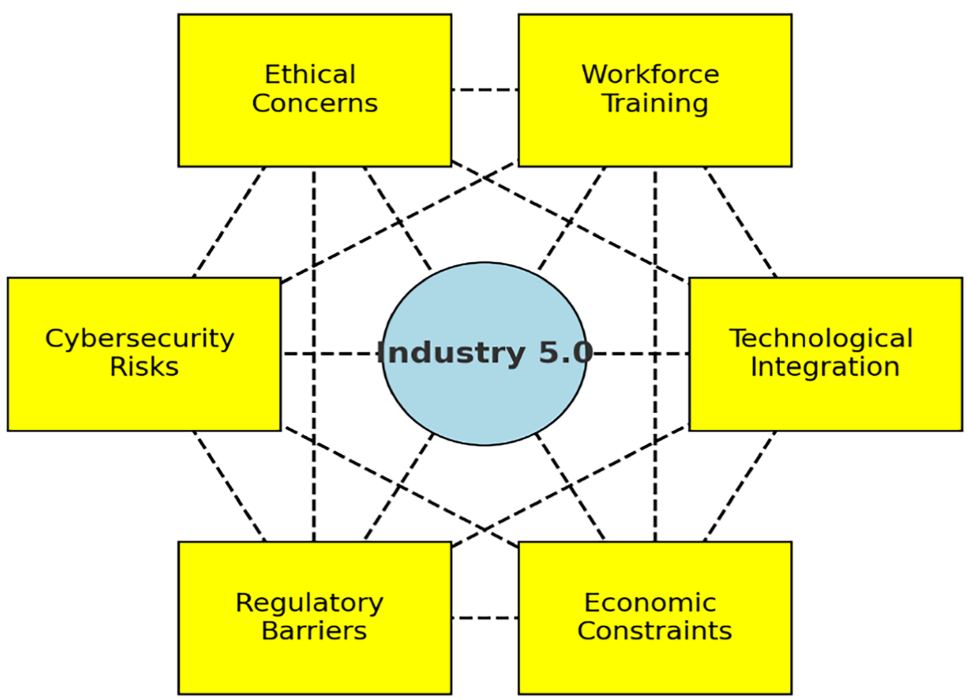

Fig. 3 Challenges of Industry 5.0 (Source: Author's creation)

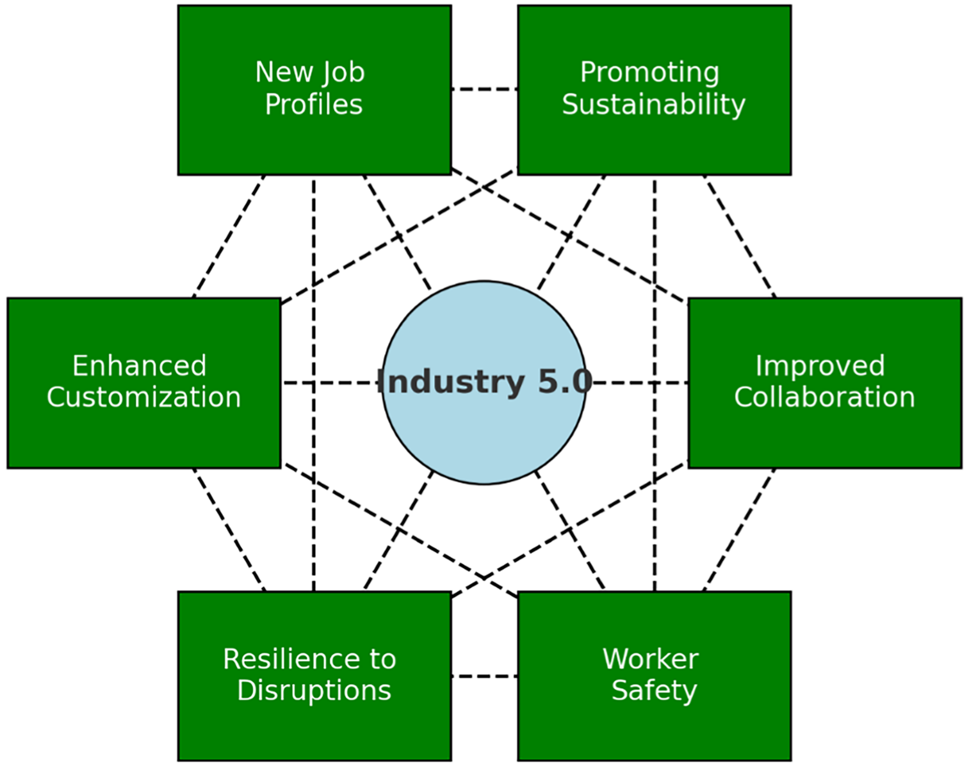

Fig. 4 Opportunities of Industry 5.0 (Source: Author's creation)

Economic and financial constraints: Implementing I5.0 technologies is resource-intensive, particularly for small and medium-sized enterprises (SMEs). The high costs associated with acquiring and maintaining advanced systems, coupled with the uncertain return on investment, may deter organizations from transitioning. Additionally, access to financial resources and government support plays a critical role in overcoming these barriers [84, 85].

3.2 Opportunities

Despite the challenges, Industry 5.0 offers transformative opportunities to redefine industrial practices as shown in Fig. 4. By prioritizing sustainability, enhancing human–machine collaboration, and promoting resilience, Industry 5.0 paves the way for a more inclusive, innovative, and sustainable industrial future [86]. These opportunities not only foster economic growth but also align with global objectives like social equity and environmental conservation, creating a balance between technological advancement and human well-being.

Enhanced customization: Industry 5.0 enables the production of personalized products at scale, known as mass customization. By leveraging human creativity and machine precision, manufacturers can deliver products tailored to individual preferences [87]. This capability fosters stronger customer loyalty, increases market competitiveness, and aligns with growing consumer demand for unique and customized solutions [88].

Promoting sustainability: I5.0 places sustainability at the core of its framework, emphasizing eco-friendly manufacturing practices. By incorporating renewable energy sources, biotechnologies, and circular economy principles, it reduces environmental impact. Sustainable production methods not only minimize waste and carbon emissions but also help industries meet global climate change goals and consumer expectations for green products [89, 90].

Improved human-machine collaboration: Collaborative robots (cobots) working alongside humans enhance operational efficiency by combining the strengths of both. While robots handle repetitive and hazardous tasks, humans focus on creative, strategic, and decision-making roles. This synergy boosts productivity, reduces workplace injuries, and fosters innovation [51, 91].

Creation of new job profiles: While some traditional roles may diminish, I5.0 is expected to create employment opportunities in emerging fields. These include AI system management, cobot programming, sustainable manufacturing practices, and bioeconomy-based industries. The demand for creative problem-solving and technical expertise provides avenues for workforce development and skill enhancement [92, 93].

Resilience to disruptions: By integrating advanced technologies, I5.0 improves the resilience of manufacturing systems. Predictive analytics and IoT enable early detection of potential disruptions, such as supply chain bottlenecks or equipment failures. This adaptability ensures smoother operations during unexpected events, including natural disasters or global crises like pandemics [94, 95].

Enhanced worker safety and well-being: Automating high-risk tasks reduces the likelihood of workplace accidents, ensuring a safer environment for employees. Moreover, by integrating human-centric values, I5.0 fosters a culture that prioritizes worker well-being, offering more meaningful roles and improving overall job satisfaction. This approach supports social equity and aligns industrial practices with human development goals [96, 97].

4 Sustainable manufacturing

Environmental, social, and economic harm can be mitigated through a paradigm shift in industrial production known as Sustainable Manufacturing (SM), often called green manufacturing [98]. This paradigm emphasizes long-term operational efficiency, waste reduction, competitiveness, and business viability within the framework of the Triple Bottom Line (TBL) sustainability—covering economic, social, and environmental dimensions [72].

Initially, SM was primarily regarded as a method to mitigate environmental damage caused by traditional manufacturing processes [99]. However, it has evolved into a holistic strategy essential for achieving global sustainability objectives. The fifth industrial revolution, Industry 5.0 (I5.0), seeks to rectify the shortcomings of Industry 4.0 (I4.0), which, despite its technological advances, often neglected environmental and social sustainability in favor of mass production. I5.0 shapes on the foundation of I4.0 by emphasizing human-centric designs, resilience, and sustainable practices.

4.1 Core principles of sustainable manufacturing in I5.0

Human-centricity: The foundation of I5.0 lies in a people-first approach. This ensures the workforce's physical and mental well-being through safe and inclusive environments while respecting workers' rights. Collaborative robots (cobots) play a crucial role in assigning repetitive or hazardous tasks to robots, allowing humans to focus on creative and strategic roles [100].

Sustainability: SM under I5.0 is deeply aligned with the principles of a circular economy. It emphasizes resource optimization by reusing materials, minimizing waste, and employing renewable energy sources such as bioenergy. These practices safeguard the environment while supporting economic growth [101, 102].

Resilience: A resilient manufacturing process under I5.0 is equipped to adapt to disruptions, whether caused by supply chain interruptions, environmental crises, or market fluctuations. This resilience is achieved by integrating predictive maintenance, flexible production systems, and robust disaster management strategies [86, 103, 104].

4.2 Technological enablers

Sustainable manufacturing under I5.0 leverages cutting-edge technologies such as [105]:

- *IoT and Big Data Analytics (BDA)* for real-time decision-making and waste reduction.
- *Artificial Intelligence (AI) and Machine Learning (ML)* for optimizing production and quality.
- *Collaborative Robotics* to enhance safety and precision.
- *Additive Manufacturing (3D printing)* for reducing material waste and enabling customization.
- *Renewable Energy Integration*, including bioenergy, for powering industrial operations.

To transition toward SM under I5.0, manufacturing enterprises must integrate sustainability into their strategic goals. This includes training workers to adapt to new technologies and roles, investing in renewable energy solutions and energy-efficient systems, developing partnerships to share expertise and overcome financial or technological barriers. The shift to sustainable manufacturing is no longer optional but imperative for addressing global challenges like climate change, resource depletion, and social inequalities. I5.0 provides a roadmap to create an inclusive, resilient, and sustainable industrial ecosystem that aligns with both business objectives and global sustainability goals [106].

5 Discussion and findings

5.1 RQ1: What defines I5.0 and what motivates SM to embrace I5.0?

Whereas I5.0 is defined by a focus on people, advanced technology integration, personalisation, sustainability, and resilience, I4.0 was defined by the use of digital technologies like the internet of things (IoT), artificial intelligence (AI), business process automation (BPA), and others to construct smart factories that are more responsive and efficient [107]. The implementation of I5.0 makes a contribution to sustainable development by fostering a balance between economic growth and the preservation of social and environmental sustainability [24]. I5.0 is not just a new paradigm that replaces or follows the I4.0 paradigm, but it is also a plan for the future of manufacturing that places an emphasis on SM and TBL.

Fundamental principles of I5.0 and aspects of sustainability for SMEs. This makes I5.0 seem like the cohabitation of value-centric I5.0 and technologically driven I4.0. Adoption of I5.0 is probably shaped by several elements. Businesses might see it as a chance to develop a competitive edge by improving responsiveness, adaptability, and efficiency of their manufacturing techniques. Important factors might include constantly rising customer expectations, including desire for personalized goods, sustainability and ethical production techniques, and ethical behavior in manufacturing. Following rules on sustainability, privacy, and data security might also inspire manufacturers to use I5.0 techniques. The long-term survival of I5.0 may be much influenced by technological developments such artificial intelligence, human–machine interfaces and improved connectivity [41]. Adoption of I5.0 may also be influenced by more general social and economic components of sustainability including changes in worker demographics, geopolitical issues and global economic reasons [17].

5.2 RQ2: How can TBL sustainability be affected by the enabling technologies supporting I5.0?

An increased human-focused approach to manufacturing is given high emphasis by I5.0, which is largely defined by I5.0 technologies; this, in turn, influences TBL sustainability [16]. Apart from promoting economic sustainability, enabling technologies greatly help to support environmental and social sustainability. Leading these changes is the IoT, a transforming tool linking devices to the internet so enabling integration, flawless communication and data sharing [42]. Critical technologies that let businesses assess vast volumes of data and derive insightful analysis are artificial intelligence and machine learning [46]. The combination of BDA with AI and ML enables the analysis and extraction of useful conclusions from the enormous data sets generated by industrial operations [49]. AR and VR technologies find application in manufacturing settings mostly for design visualization, maintenance, and training [58]. These technologies improve manufacturing adaptability by facilitating instant decision-making, proactive maintenance, and streamlined process management [108].

Collaborative robots, often known as cobots, have increased industrial robotics and automation considerably and might soon be a necessary part of manufacturing lines [38]. The manufacturing process has been totally changed by additive manufacture. Its capacity to generate sophisticated and customized components reduces waste and speeds up prototype development, so supporting a more sustainable and agile production environment [109]. Not only do these technological advancements make the workplace safer for employees, but they also increase precision and efficacy.

5.3 RQ3: Among different industries, what possibilities and related difficulties does I5.0 present?

The manufacturing industry has been actively adopting the smart manufacturing model in recent years [17]. Like any revolution, though, this one provides both fresh opportunities and related difficulties. Consequently, it is crucial to proactively tackle the obstacles while making the most of the potential given by I5.0. This is particularly true in any developing country, where manufacturing not only accounts for the lion's share of GDP but also provides numerous employment opportunities through micro, small, and medium-sized enterprises (MSMEs).

Industry 5.0 introduces transformative opportunities across multiple sectors, fostering innovation, efficiency, and sustainability. In the healthcare sector, it enables customized implants, enhances remote healthcare services, and improves precision in surgeries, leveraging AI and IoT. In manufacturing, Industry 5.0 promotes high productivity, innovation, and energy-efficient systems, supporting flexible and sustainable production. The education sector benefits from enhanced creativity, visualization, and interactive learning approaches, allowing for personalized and immersive educational experiences [4].

Smart transportation and traffic management are revolutionized through human-AI collaboration, 5G networks, and real-time data monitoring, leading to more efficient and safe mobility solutions. In supply chain management, smart replenishment systems help reduce inventory costs and eliminate stock-outs, ensuring seamless logistics. Disaster management benefits from real-time monitoring and proactive resource operations, enhancing resilience in crisis situations. Smart cities leverage Industry 5.0 for efficient energy use, waste management, and real-time urban monitoring, creating a more sustainable and responsive urban environment. In agriculture, precision farming using IoT and AI optimizes resource usage and reduces waste, improving yield and environmental impact. The energy sector benefits from renewable energy integration and optimized grid management, ensuring sustainability and efficiency. Finally, in the retail sector, Industry 5.0 enhances personalized shopping experiences, predictive demand management, and faster delivery systems, transforming customer engagement and supply chain operations [95].

Despite its transformative potential, Industry 5.0 presents several challenges. A major hurdle is the need for a robust digital infrastructure, including high-speed internet connectivity, to enable seamless operations. Work polarization remains a critical concern, as Industry 5.0-driven automation may create disparities between highly skilled and low-skilled workers [31]. Regulatory compliance poses another challenge, as policies struggle to keep pace with rapidly evolving technologies, requiring clear guidelines for ethical AI and automation. Skill development is essential to equip the workforce with the competencies needed to handle advanced technologies, necessitating extensive training programs. The high initial cost of implementing Industry 5.0 technologies poses financial constraints, particularly for small and medium-sized enterprises (SMEs), limiting widespread adoption [110]. Cybersecurity risks also emerge due to increased connectivity, emphasizing the need for strong data protection measures to prevent cyber threats. Scalability is another challenge, as implementing Industry 5.0 solutions on a large scale requires substantial investment and adaptation. Ensuring resilience and adaptability in the face of supply chain disruptions and external crises is also crucial. Privacy concerns related to increased data monitoring and sharing need to be addressed to maintain user trust. Lastly, human–robot collaboration introduces safety risks in highly automated environments, necessitating strict guidelines to ensure worker safety and effective integration [95].

5.4 RQ5: building a collaborative ecosystem for sustainable manufacturing with industry 5.0

Industry 5.0 transcends technological advancements by emphasizing collaboration among diverse stakeholders to achieve sustainable manufacturing. This section explores a framework that fosters a collaborative ecosystem involving governments, industry players, academia, and communities to facilitate a smooth transition to Industry 5.0.

5.4.1 Framework components

The proposed collaborative ecosystem framework incorporates the following elements:

Stakeholder engagement: Stakeholder engagement is a cornerstone of the collaborative ecosystem framework. Governments play a crucial role by designing policies and incentives that encourage the adoption of Industry 5.0 principles, such as providing tax benefits for using renewable energy or implementing human-centric designs. Industries, on the other hand, are encouraged to collaborate by sharing resources and expertise to overcome barriers like high costs and skill gaps. This collaboration can be formalized through consortia focused on knowledge sharing and co-innovation. Academia and research institutions contribute by developing advanced training modules and conducting research to address skill development and emerging challenges. Finally, communities are integrated into the process by fostering social acceptance and aligning industrial goals with societal needs through active participation and dialogue.

Shared digital infrastructure: The establishment of shared digital infrastructure is vital for enabling real-time data sharing and streamlined operations among stakeholders. Interconnected platforms facilitate seamless communication across the manufacturing ecosystem, enhancing efficiency and reducing redundancies. These platforms also ensure the availability of open-access research and innovation repositories, which can drive continuous improvement and collaboration. By leveraging a shared digital ecosystem, stakeholders can integrate advanced technologies like IoT and AI while maintaining transparency and interoperability across sectors.

Circular Economy Integration: Circular economy integration focuses on reducing waste and optimizing resource usage within the manufacturing ecosystem. This involves implementing systems for reuse, recycling, and remanufacturing to create sustainable production cycles. Industries are encouraged to share responsibility for managing environmental impacts, promoting practices that align with the principles of sustainability. By embedding circular economy principles, the framework aims to minimize ecological footprints while fostering long-term economic and environmental benefits.

Human-centric innovation: Human-centric innovation emphasizes the importance of designing workplaces and processes that prioritize the well-being and inclusivity of employees. This includes creating environments that integrate collaborative robots (cobots) to handle repetitive or hazardous tasks, allowing employees to focus on creative and strategic roles. Forums for employees to voice concerns and contribute to decision-making processes are also essential. By placing humans at the center of innovation, the framework ensures that technological advancements are aligned with social and ethical considerations.

Skill development and training: Skill development and training are critical to equipping the workforce with the capabilities needed for Industry 5.0. This involves launching cross-industry training programs that focus on practical applications of enabling technologies such as AI, IoT, and robotics. Additionally, vocational education tailored to the specific demands of Industry 5.0 ensures that workers are prepared for new roles and challenges. By addressing skill gaps and fostering continuous learning, the framework supports a smooth and inclusive transition to advanced manufacturing practices.

5.4.2 Implementation steps

1. Technology-Human Integration: Industry 5.0 enhances human–machine collaboration by integrating AI, collaborative robots (cobots), and human-digital twins (HDTs). Unlike traditional automation, it augments human capabilities rather than replacing them. AI-driven collaborative intelligence enables better decision-making, while cobots assist in precision tasks. HDTs monitor worker well-being and ergonomics, ensuring a safer, more adaptive work environment. This synergy boosts productivity, efficiency, and workforce satisfaction.

2. Ethical and Regulatory Compliance: With increased automation, ethical AI, data privacy, and legal frameworks are crucial. AI-driven decision-making risks bias and transparency issues, necessitating explainable AI (XAI) to ensure fairness. The use of HDTs and biometric data demands strong regulatory oversight for privacy and cybersecurity. Governments and industries must establish clear guidelines for human–robot collaboration, ensuring ethical AI adoption and workplace trust.

3. Sustainability-Driven Smart Manufacturing: Industry 5.0 integrates circular economy principles, renewable energy, and AI-driven efficiency to address Industry 4.0's environmental impact. Smart monitoring, AI-powered material recycling, and predictive energy management reduce waste and enhance sustainability. By embedding eco-friendly automation and resource optimization, Industry 5.0 ensures long-term environmental and economic benefits.

4. Workforce Reskilling & Adaptation: A skilled workforce is essential for Industry 5.0's human-AI synergy. AI-assisted training, immersive simulations, and interdisciplinary learning help workers adapt to evolving roles. Reskilling initiatives reduce job displacement fears, enabling a smooth transition into AI-enhanced environments while boosting workforce confidence and productivity.

5. Real-Time Feedback & Continuous Improvement: Industry 5.0 relies on dynamic AI-driven feedback to refine human–machine interactions and optimize workflows. HDTs track worker conditions, predictive analytics improve adaptability, and self-learning AI systems enhance efficiency. This real-time optimization leads to higher productivity, resilience, and a more responsive industrial ecosystem.

5.4.3 Benefits of the proposed framework

The collaborative ecosystem framework provides numerous benefits by aligning stakeholders' efforts and leveraging shared resources. Economically, it reduces redundancy and operational costs while fostering innovation across industries. Environmentally, it accelerates the adoption of sustainable practices, facilitating a shift toward circular economies and reducing the overall ecological footprint. Socially, it promotes inclusivity and skill development, ensuring equitable participation in decision-making processes. By balancing economic, social, and environmental goals, the framework lays the foundation for a resilient, sustainable manufacturing ecosystem.

5.4.4 Ethical AI, bias in automation, and regulatory frameworks in industry 5.0

As Industry 5.0 integrates collaborative robotics, human digital twins (HDTs), and AI-driven decision-making, it is essential to address the ethical implications, potential biases in automation, and regulatory challenges associated with these technologies. While Industry 5.0 prioritizes human-centricity, ensuring fairness, transparency, and accountability in AI-driven processes remains a key concern.

5.4.4.1 Ethical AI and bias in automation AI systems used in predictive maintenance, workforce allocation, and production planning can inherit biases from training data, leading to unintended discrimination or unequal decision-making. Bias in AI-driven recruitment, task assignments, or performance evaluations may negatively impact diversity and inclusion in the workforce [55]. Ensuring ethical AI requires transparency, explainability, and robust validation mechanisms to detect and correct biases before deployment.

5.4.4.2 Regulatory challenges in human–robot collaboration The increasing presence of collaborative robots (Cobots) and HDTs in manufacturing raises concerns about safety, liability, and legal accountability. Regulatory frameworks must address: workplace safety standards to ensure human workers are protected in AI-assisted environments, data privacy laws to prevent misuse of worker biometric data collected through HDTs, liability in AI-driven decision-making, ensuring that organizations remain accountable for automated errors or ethical lapses.

Currently, regulatory policies struggle to keep pace with technological advancements, leading to inconsistencies in AI governance and cobot deployment across industries [54]. Establishing international AI ethics guidelines and workplace safety policies is crucial to prevent misuse while promoting fair and transparent human–robot collaboration.

5.4.4.3 Responsible AI in collaborative workspaces For AI to be effectively integrated into smart manufacturing and human-centric production, it must follow principles of fairness, accountability, and human oversight. Collaborative AI systems should be auditable, interpretable, and aligned with ethical AI guidelines to ensure trust in human–machine interactions [55]. Adopting explainable AI models, human-in-the-loop validation, and unbiased datasets can help mitigate ethical risks while maintaining productivity and innovation in Industry 5.0.

By addressing ethical AI, regulatory challenges, and bias in automation, Industry 5.0 can ensure a fair, safe, and human-centric future while maintaining its sustainability and efficiency goals. Future research should focus on developing standardized AI ethics frameworks and regulatory policies to guide responsible technological adoption.

6 Implications

This study provides critical theoretical and practical insights into the transition from Industry 4.0 to Industry 5.0, emphasizing its transformative potential for sustainable manufacturing. The findings highlight the strategic role of Industry 5.0 in fostering a human-centric, technology-enabled, and sustainability-oriented industrial paradigm, addressing critical gaps left by its predecessor.

6.1 Theoretical implications

The research advances the theoretical understanding of Industry 5.0 by distinguishing its human-centric and sustainability-driven approach from the automation-focused paradigm of Industry 4.0. It extends existing literature on sustainable manufacturing by integrating the principles of the Triple Bottom Line (TBL), which balances economic, social, and environmental dimensions. The theoretical contribution lies in conceptualizing Industry 5.0 not just as an evolutionary step but as a value-driven industrial model that prioritizes resilience, personalization, and ecological sustainability.

Moreover, the study deepens the discourse on human–machine collaboration, positioning humans as central actors in decision-making and creative processes while leveraging advanced technologies for efficiency and safety. It also expands the theoretical framework for sustainable manufacturing by embedding concepts like the circular economy, renewable energy integration, and waste minimization into the Industry 5.0 ecosystem. These insights contribute to a holistic framework for researchers exploring the convergence of technology, human agency, and sustainability in industrial contexts.

6.2 Practical implications

Practically, the findings offer actionable insights for manufacturers, industry leaders, and organizational strategists. Industry 5.0 introduces collaborative robots (cobots), advanced AI, and IoT as key enablers for human-centric and efficient production systems. These technologies enable enhanced mass customization, improved safety, and optimized resource usage, presenting opportunities to address consumer demand for personalized products while reducing environmental impact.

For small and medium-sized enterprises (SMEs), the study highlights pathways to adopt Industry 5.0 technologies despite resource constraints. Shared digital infrastructure, public–private partnerships, and targeted training initiatives are emphasized as mechanisms to democratize access to cutting-edge innovations. Moreover, the integration of predictive analytics and blockchain offers practical solutions for enhancing supply chain resilience, ensuring adaptive responses to global disruptions. The focus on workforce development underscores the necessity of retraining employees to work with advanced technologies. The findings emphasize the importance of cultivating digital and cognitive skills while aligning organizational culture with Industry 5.0 principles. This approach fosters innovation, operational flexibility, and long-term competitiveness.

6.3 Implications for policymakers

For policymakers, this study underscores the urgency of developing supportive frameworks to facilitate the transition to Industry 5.0. Governments must prioritize investments in digital infrastructure, including high-speed internet and IoT networks, to enable seamless technological integration. Additionally, policy interventions should include financial incentives such as tax breaks and subsidies for adopting renewable energy solutions, implementing sustainable practices, and fostering human-centric innovation. Workforce upskilling must be a central focus of policy frameworks. Initiatives such as vocational training programs, industry-academia collaborations, and certification courses in advanced manufacturing technologies can address skill gaps and promote inclusive industrial growth. Policymakers should also establish guidelines and regulatory frameworks to address ethical concerns related to data privacy, AI accountability, and workplace equity.

Lastly, the findings encourage the creation of multistakeholder ecosystems involving governments, industries, academic institutions, and communities. Such ecosystems can drive knowledge sharing, resource pooling, and co-innovation, ensuring that the transition to Industry 5.0 aligns with regional economic priorities and global sustainability goals.

6.4 Empirical implications

The findings of this study provide a strong foundation for empirical research, offering insights that can be further tested through quantitative and qualitative methodologies. To facilitate empirical validation, we propose a set of testable hypotheses and a refined Industry 5.0 framework that outlines specific implementation steps.

7 Framework for empirical testing

Industry 5.0 presents a complex interplay between human-centric automation, sustainability, and technological advancement. To assess these dynamics in real-world applications, we propose the following hypotheses for future empirical research:

H1: The integration of collaborative robotics (cobots) enhances worker efficiency and safety while maintaining high production flexibility.

H2: Human-digital twins (HDTs) improve worker well-being and adaptive manufacturing processes, leading to higher productivity.

H3: AI-driven decision-making in Industry 5.0 reduces bias and increases trust when combined with human oversight.

H4: Firms adopting Industry 5.0 principles experience greater sustainability benefits through circular economy integration and reduced resource consumption.

These hypotheses provide a structured basis for future experimental studies, industrial case analyses, and workforce impact assessments.

8 Limitations and future research directions

This study provides valuable insights into Industry 5.0, but several limitations must be acknowledged. The case study selection is not exhaustive, as Industry 5.0 is still evolving, and further research should explore its adoption in developing economies and non-manufacturing sectors. Additionally, the study relies on existing literature and theoretical analysis, lacking empirical validation through field trials or industrial case studies. The discussion on ethical AI and regulatory challenges is based on current policies, which continue to evolve, requiring ongoing research into their real-world implications. While the study focuses on manufacturing, workforce, and sustainability, Industry 5.0's impact on healthcare, finance, and other industries remains underexplored. Moreover, economic and technological barriers—such as high implementation costs and workforce training requirements—need further analysis, particularly for small and medium-sized enterprises (SMEs).

Future research should focus on empirical validation through case studies and sector-specific analyses to uncover unique challenges and opportunities. Longitudinal studies could assess the evolving impacts of Industry 5.0 technologies on sustainability and resilience. In-depth exploration of policy frameworks and ethical dimensions is essential for guiding equitable adoption. Expanding research to include emerging technologies like quantum computing and bioengineering could provide further insights into their integration with Industry 5.0 principles. Addressing these gaps will strengthen the theoretical and practical foundation for advancing human-centric and sustainable industrial practices.

9 Conclusion

This study provides a comprehensive exploration of Industry 5.0, emphasizing its transformative potential to redefine sustainable manufacturing through a human-centric, technologically advanced, and sustainability-oriented framework. By addressing the limitations of Industry 4.0—such as excessive automation and insufficient consideration of social and environmental dimensions—Industry 5.0 emerges as a paradigm that balances technological innovation with ethical and ecological responsibility. The integration of enabling technologies such as collaborative robotics, artificial intelligence, and the Internet of Things underlines Industry 5.0's capacity to enhance productivity while promoting mass customization and reducing environmental impact. The study underscores the importance of the Triple Bottom Line (TBL) approach, which prioritizes economic growth, environmental preservation, and social equity. These principles are especially relevant for emerging economies, where Industry 5.0 presents opportunities to address socio-economic challenges and position nations as leaders in sustainable industrial practices.

While the findings offer significant theoretical and practical contributions, they also highlight critical challenges, including skill gaps, technological integration barriers, and ethical concerns. Addressing these issues requires robust policy frameworks, stakeholder collaboration, and continuous workforce development. Policymakers and industry leaders must foster an inclusive ecosystem that aligns technological advancements with societal needs and ecological sustainability. In conclusion, Industry 5.0 represents more than a technological evolution; it is a value-driven revolution that places humans and sustainability at the center of industrial progress. By leveraging advanced technologies in harmony with ethical and sustainable practices, Industry 5.0 lays the foundation for a resilient, inclusive, and environmentally responsible future. Further research and cross-sector collaboration will be instrumental in realizing this vision and ensuring its long-term success.

9.1 Key contributions of this study

This study makes several significant contributions to the understanding and implementation of Industry 5.0, bridging the gap between theoretical advancements and practical applications.

A comprehensive analysis of industry 5.0: This paper provides a structured and cross-referenced review of Industry 5.0, emphasizing its technological advancements, sustainability impact, and ethical considerations. By synthesizing insights from multiple sources, this study offers a balanced discussion on both challenges and opportunities.

A refined implementation framework: We propose a detailed Industry 5.0 framework, outlining key steps in human–machine collaboration, ethical AI governance, sustainability-driven manufacturing, workforce reskilling, and real-time optimization. This structured approach ensures that Industry 5.0 adoption is efficient, responsible, and adaptable.

Empirical applicability through hypothesis development: To enable future empirical research, we formulate testable hypotheses related to the impact of collaborative robotics, human-digital twins, AI-driven decision-making, and sustainability efforts. These hypotheses provide a foundation for experimental validation and industrial case studies, ensuring the study's real-world relevance.

A structured approach to ethical AI and regulatory challenges: The study highlights key ethical concerns in AI-driven automation and proposes strategies for explainable AI (XAI), bias mitigation, and regulatory compliance. By addressing current gaps in governance, this research contributes to the development of responsible AI policies for Industry 5.0 adoption.

By integrating these contributions, this study extends existing literature and provides a practical roadmap for researchers, policymakers, and industries aiming to transition towards a human-centric and sustainable industrial future. Future research should focus on empirical validation and policy development to further refine Industry 5.0 implementation strategies.

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References

1. Shabur MdA. A comprehensive review on the impact of Industry 4.0 on the development of a sustainable environment. Discov Sustain. 2024;5(1):97. https://doi.org/10.1007/s43621-024-00290-7.
2. Sarkar BD, Shardeo V, Dwivedi A, Pamucar D, Digital transition from industry 4.0 to industry 5.0 in smart manufacturing: A framework for sustainable future, vol. 78. Elsevier, 2024, p. 102649. [Online]. https://www.sciencedirect.com/science/article/pii/S0160791X24001970. Accessed 31 Dec 2024.

3. Grabowska S, Saniuk S, Gajdzik B. Industry 5.0: improving humanization and sustainability of Industry 4.0. Scientometrics. 2022;127(6):3117–44. https://doi.org/10.1007/s11192-022-04370-1.
4. Maddikunta PKR, et al. Industry 5.0: A survey on enabling technologies and potential applications, vol. 26. Elsevier, 2022, p. 100257. [Online]. https://www.sciencedirect.com/science/article/pii/S2452414X21000558. Accessed 31 Dec 2024.
5. Jefroy N, Azarian M, Yu H, Moving from Industry 4.0 to Industry 5.0: what are the implications for smart logistics?, vol. 6, no. 2. MDPI, 2022, p. 26. https://www.mdpi.com/2305-6290/6/2/26. Accessed 31 Dec 2024.
6. Xu X, Lu Y, Vogel-Heuser B, Wang L, Industry 4.0 and Industry 5.0—inception, conception and perception, vol. 61. Elsevier, 2021, pp. 530–535. https://www.sciencedirect.com/science/article/pii/S0278612521002119. Accessed 31 Dec 2024.
7. Xu LD, Xu EL, Li L, Industry 40: state of the art and future trends. Int J Prod Res, vol. 56, no. 8, Art. no. 8, 2018.
8. Vinitha K, Prabhu RA, Bhaskar R, Hariharan R, Review on industrial mathematics and materials at Industry 1.0 to Industry 4.0, vol. 33. Elsevier, 2020, pp. 3956–3960. https://www.sciencedirect.com/science/article/pii/S2214785320348045. Accessed 31 Dec 2024.
9. Yin Y, Stecke KE, Li D. The evolution of production systems from Industry 2.0 through Industry 4.0. Int J Prod Res. 2018;56(1–2):848–61. https://doi.org/10.1080/00207543.2017.1403664.
10. Shabur A, Hridoy MW, Rahman KA. The investigation of challenges of implementing industry 4.0 in Bangladesh. Acad Lett. 2021. https://doi.org/10.20935/AL775.
11. Zheng T, Ardolino M, Bacchetti A, Perona M. The applications of Industry 4.0 technologies in manufacturing context: a systematic literature review. Int J Prod Res. 2021;59(6):1922–54. https://doi.org/10.1080/00207543.2020.1824085.
12. Shabur MdA, Rahman KA, Siddiki MdR. Evaluating the difficulties and potential responses to implement Industry 40 in Bangladesh's steel sector. J Eng Appl Sci. 2023;70(1):158. https://doi.org/10.1186/s44147-023-00336-z.
13. Oztemel E, Gursev S. Literature review of industry 4.0 and related technologies. J Intell Manuf. 2020;31(1):127–82. https://doi.org/10.1007/s10845-018-1433-8.
14. Chiarini A. Industry 4.0 technologies in the manufacturing sector: are we sure they are all relevant for environmental performance? Bus Strat Env. 2021;30(7):3194–207. https://doi.org/10.1002/bse.2797.
15. D. Zuehlke, *SmartFactory—Towards a factory-of-things*, vol. 34, no. 1. Elsevier, 2010, pp. 129–138. [Online]. https://www.sciencedirect.com/science/article/pii/S1367578810000143. Accessed 31 Dec 2024.
16. Ivanov D, Dolgui A, Sokolov B, Werner F, Ivanova M. A dynamic model and an algorithm for short-term supply chain scheduling in the smart factory industry 4.0. Int J Prod Res. 2016;54(2):386–402. https://doi.org/10.1080/00207543.2014.999958.
17. Pasi BN, Mahajan SK, Rane SB. The current sustainability scenario of Industry 4.0 enabling technologies in Indian manufacturing industries. Int J Product Perform Manag. 2021;70(5):1017–48.
18. Shabur MdA, Ali MdF, Alam MdM. Analysis of the barriers and possible approaches for adopting Industry 40 in the fertilizer sector of Bangladesh. Discov Appl Sci. 2024;6(7):369. https://doi.org/10.1007/s42452-024-06074-y.
19. Jayashree S, Reza MNH, Malarvizhi CAN, Mohiuddin M, Industry 4.0 implementation and Triple Bottom Line sustainability: An empirical study on small and medium manufacturing firms, vol. 7, no. 8. Elsevier, 2021. [Online]. https://www.cell.com/heliyon/fulltext/S2405-8440(21)01856-9. Accessed 31 Dec 2024.
20. Narkhede G, Mahajan S, Narkhede R, Chaudhari T. Significance of Industry 4.0 technologies in major work functions of manufacturing for sustainable development of small and medium-sized enterprises. Bus Strat Dev. 2024;7(1):e325. https://doi.org/10.1002/bsd2.325.
21. Sreedharan VR, Unnikrishnan A. Moving towards industry 4.0: a systematic review. Int J Pure Appl Math. 2017;117(20):929–36.
22. Ghobakhloo M, Iranmanesh M, Tseng M-L, Grybauskas A, Stefanini A, Amran A. Behind the definition of Industry 5.0: a systematic review of technologies, principles, components, and values. J Ind Prod Eng. 2023;40(6):432–47. https://doi.org/10.1080/21681015.2023.2216701.
23. Grybauskas A, Stefanini A, Ghobakhloo M, Social sustainability in the age of digitalization: a systematic literature Review on the social implications of industry 4.0, vol. 70. Elsevier, 2022, p. 101997. [Online]. https://www.sciencedirect.com/science/article/pii/S0160791X22001385 Accessed 31 Dec 2024.
24. Borchardt M, Pereira GM, Milan GS, Scavarda AR, Nogueira EO, Poltosi LC. Industry 5.0 beyond technology: an analysis through the lens of business and operations management literature. Organizacija. 2022;55(4):305–21. https://doi.org/10.2478/orga-2022-0020.
25. Gladysz B, Tran T, Romero D, van Erp T, Abonyi J, Ruppert T, Current development on the Operator 4.0 and transition towards the Operator 5.0: A systematic literature review in light of Industry 5.0, vol. 70. Elsevier, 2023, pp. 160–185. [Online]. https://www.sciencedirect.com/science/article/pii/S0278612523001401. Accessed 31 Dec 2024.
26. Tranfield D, Denyer D, Smart P. Towards a methodology for developing evidence-informed management knowledge by means of systematic review. British J of Management. 2003;14(3):207–22. https://doi.org/10.1111/1467-8551.00375.
27. Mengist W, Soromessa T, Legese G, Method for conducting systematic literature review and meta-analysis for environmental science research, vol. 7. Elsevier, 2020, p. 100777. [Online]. https://www.sciencedirect.com/science/article/pii/S221501611930353X. Accessed 31 Dec 2024.
28. Page MJ, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews, vol. 372. British Medical Journal Publishing Group, 2021. [Online]. https://www.bmj.com/content/372/bmj.n71.short. Accessed 31 Dec 2024.
29. Snyder H, Literature review as a research methodology: An overview and guidelines, vol. 104. Elsevier, 2019, pp. 333–339. [Online]. https://www.sciencedirect.com/science/article/pii/S0148296319304564. Accessed 31 Dec 2024.
30. Kitchenham B, Brereton P, Zhi L, Budgen D, Burn A, Repeatability of systematic literature reviews," In: 15th Annual Conference on Evaluation & Assessment in Software Engineering (EASE 2011), Durham, UK: IET, 2011, pp. 46–55. https://doi.org/10.1049/ic.2011.0006.
31. Baranauskas G, Mass personalization vs. mass customization: Finding variance in semantical meaning and practical implementation between sectors, In: Social transformations in contemporary society (STICS 2019): proceedings of an annual international conference for young researchers. Vilnius: Mykolas Romeris university, 2019, no. 7., 2019. [Online]. https://cris.mruni.eu/cris/bitstream/007/15843/1/STICS_2019_7_6-15.pdf?sequence=1. Accessed 05 Jan 2025.
32. Dieste M, Orzes G, Culot G, Sartor M, Nassimbeni G. The 'dark side' of Industry 4.0: How can technology be made more sustainable? IJOPM. 2024;44(5):900–33. https://doi.org/10.1108/IJOPM-11-2022-0754.

33. Pech M, Vaněček D, Barriers of new technology introduction and disadvantages of industry 4.0 for industrial enterprises, vol. 17, no. 1. 2022, pp. 197–206. [Online]. https://www.aseestant.ceon.rs/index.php/sjm/article/view/30453. Accessed 05 Jan 2025.
34. Miraz MH, Hasan MT, Sumi FR, Sarkar S, Hossain MA, Industry 5.0: The Integration of Modern Technologies, In: *Machine Vision for Industry 4.0*, CRC Press, 2022, pp. 285–300. [Online]. https://www.taylorfrancis.com/chapters/edit/https://doi.org/10.1201/9781003122401-14/industry-5-0-mahadi-hasan-miraz-mohammad-tariq-hasan-farhana-rahman-sumi-shumi-sarkar-mohammad-amzad-hossain. Accessed 05 Jan 2025.
35. Golovianko M, Terziyan V, Branytskyi V, Malyk D. Industry 4.0 vs. industry 5.0: co-existence, transition, or a hybrid. Procedia Comput Sci. 2023;217:102–13.
36. Rožanec JM, et al. Human-centric artificial intelligence architecture for industry 5.0 applications. Int J Prod Res. 2023;61(20):6847–72. https://doi.org/10.1080/00207543.2022.2138611.
37. Demir KA, Cicibaş H, The next industrial revolution: industry 5.0 and discussions on industry 4.0, In: 4th International Management Information Systems Conference "Industry, 2019, pp. 17–20.
38. Evjemo LD, Gjerstad T, Grøtli EI, Sziebig G. Trends in smart manufacturing: role of humans and industrial robots in smart factories. Curr Robot Rep. 2020;1(2):35–41. https://doi.org/10.1007/s43154-020-00006-5.
39. Dalenogare LS, Benitez GB, Ayala NF, Frank AG. The expected contribution of Industry 4.0 technologies for industrial performance. Int J Prod Econ. 2018;204:383–94. https://doi.org/10.1016/j.ijpe.2018.08.019.
40. Moeuf A, Pellerin R, Lamouri S, Tamayo-Giraldo S, Barbaray R. The industrial management of SMEs in the era of Industry 4.0. Int J Prod Res. 2018;56(3):1118–36. https://doi.org/10.1080/00207543.2017.1372647.
41. Bednar PM, Welch C. Socio-technical perspectives on smart working: creating meaningful and sustainable systems. Inf Syst Front. 2020;22(2):281–98. https://doi.org/10.1007/s10796-019-09921-1.
42. Belhadi A, Zkik K, Cherrafi A, Shari MY. Understanding big data analytics for manufacturing processes: insights from literature review and multiple case studies. Comput Indus Eng. 2019;137:106099.
43. Sarker MdR, Islam M, Marma UAS, Alam MdM, Shabur MdA, Rahman MS. Circular economy strategies: a fuzzy DEMATEL decision framework for the fast fashion footwear manufacture. Discov Sustain. 2024;5(1):260. https://doi.org/10.1007/s43621-024-00484-z.
44. Zhong RY, Newman ST, Huang GQ, Lan S, Big Data for supply chain management in the service and manufacturing sectors: Challenges, opportunities, and future perspectives, vol. 101. Elsevier, 2016, pp. 572–591. [Online]. https://www.sciencedirect.com/science/article/pii/S0360835216302388. Accessed 05 Jan 2025.
45. Li Z, Wang Y, Wang K-S. Intelligent predictive maintenance for fault diagnosis and prognosis in machine centers: Industry 4.0 scenario. Adv Manuf. 2017;5(4):377–87. https://doi.org/10.1007/s40436-017-0203-8.
46. Cioffi M, Travaglioni G, Piscitell, Petrillo A, De Felice F, Artificial intelligence and machine learning applications in smart production: Progress, trends, and directions, vol. 12, no. 2. MDPI, 2020, p. 492. [Online]. https://www.mdpi.com/2071-1050/12/2/492. Accessed 05 Jan 2025.
47. Çınar ZM, AbdussalamNuhu A, Zeeshan Q, Korhan O, Asmael M, Safaei B. Machine learning in predictive maintenance towards sustainable smart manufacturing in industry 40. Sustainability. 2020;12(19):8211.
48. Wang J, Xu C, Zhang J, Zhong R. Big data analytics for intelligent manufacturing systems: a review. J Manuf Syst. 2022;62:738–52. https://doi.org/10.1016/j.jmsy.2021.03.005.
49. Maheshwari S, Gautam P, Jaggi CK. Role of big data analytics in supply chain management: current trends and future perspectives. Int J Prod Res. 2021;59(6):1875–900. https://doi.org/10.1080/00207543.2020.1793011.
50. Chehbi-Gamoura S, Derrouiche R, Damand D, Barth M. Insights from big Data Analytics in supply chain management: an all-inclusive literature review using the SCOR model. Prod Plan Control. 2020;31(5):355–82. https://doi.org/10.1080/09537287.2019.1639839.
51. Keshvarparast A, Battini D, Battaia O, Pirayesh A. Collaborative robots in manufacturing and assembly systems: literature review and future research agenda. J Intell Manuf. 2024;35(5):2065–118. https://doi.org/10.1007/s10845-023-02137-w.
52. Montini E, et al. Collaborative robotics: a survey from literature and practitioners perspectives. J Intell Robot Syst. 2024;110(3):117. https://doi.org/10.1007/s10846-024-02141-z.
53. Wang B, et al. Human digital twin in the context of industry 5.0. Robot Comp Integr Manuact. 2024;85: 102626.
54. Cutrona V, Bonomi N, Montini E, Ruppert T, Delinavelli G, Pedrazzoli P. Extending factory digital Twins through human characterisation in Asset Administration Shell. Int J Comput Integr Manuf. 2024;37(10–11):1214–31. https://doi.org/10.1080/0951192X.2023.2278108.
55. E. Schleiger, C. Mason, C. Naughtin, A. Reeson, and C. Paris, *Collaborative Intelligence: A scoping review of current applications*, vol. 38, no. 1. Taylor & Francis, 2024, p. 2327890. [Online]. https://www.tandfonline.com/doi/abs/https://doi.org/10.1080/08839514.2024.2327890%4010.1080/tfocoll.2024.0.issue-Artificial-Intelligence-Applications.
56. Zafar MH, Langås EF, Sanfilippo F, Exploring the synergies between collaborative robotics, digital twins, augmentation, and industry 5.0 for smart manufacturing: a state-of-the-art review, vol. 89. Elsevier, 2024, p. 102769. https://www.sciencedirect.com/science/article/pii/S0736584524000553. Accessed 23 Feb 2025.
57. Doyle-Kent M, Kopacek P, Adoption of collaborative robotics in industry 5.0. An Irish industry case study, vol. 54, no. 13. Elsevier, 2021, pp. 413–418. [Online]. https://www.sciencedirect.com/science/article/pii/S2405896321019200. Accessed 05 Feb 2025.
58. Kamble SS, Gunasekaran A, Ghadge A, Raut R. A performance measurement system for industry 4.0 enabled smart manufacturing system in SMMEs-a review and empirical investigation. Int J Prod Econ. 2020;229: 107853.
59. Matana G, Simon A, GodinhoFilho M, Helleno A. Method to assess the adherence of internal logistics equipment to the concept of CPS for industry 40. Int jJ Prod Econ. 2020;228:107845.
60. Kusiak A. Smart manufacturing. Int J Prod Res. 2018;56(1–2):508–17. https://doi.org/10.1080/00207543.2017.1351644.
61. Soh C, Connolly D. New frontiers of profit and risk: the fourth industrial revolution's impact on business and human rights. New Political Economy. 2021;26(1):168–85. https://doi.org/10.1080/13563467.2020.1723514.
62. Skobelev PO, Borovik SY, On the way from Industry 4.0 to Industry 50: From digital manufacturing to digital society, Industry 40, 2017;2(6): 307–311, 2017.
63. Mian SH, Salah B, Ameen W, Moiduddin K, Alkhalefah H. Adapting universities for sustainability education in industry 4.0: channel of challenges and opportunities. Sustainability. 2020;12(15):6100.

64. Margherita EG, Braccini AM. Industry 4.0 technologies in flexible manufacturing for sustainable organizational value: reflections from a multiple case study of italian manufacturers. Inf Syst Front. 2023;25(3):995–1016. https://doi.org/10.1007/s10796-020-10047-y.
65. Strandhagen JW, Alfnes E, Strandhagen JO, Vallandingham LR. The fit of Industry 4.0 applications in manufacturing logistics: a multiple case study. Adv Manuf. 2017;5(4):344–58. https://doi.org/10.1007/s40436-017-0200-y.
66. Müller J, Enabling technologies for industry 5.0: results of a workshop with europe's technology leaders, Directorate-General for Research and Innovation, 2020.
67. Nahavandi S. Industry 5.0—a human-centric solution. Sustainability. 2019;11(16):4371.
68. Adel A. Future of industry 5.0 in society: human-centric solutions, challenges and prospective research areas. J Cloud Comp. 2022;11(1):40. https://doi.org/10.1186/s13677-022-00314-5.
69. Ghobakhloo M, Iranmanesh M, Mubarak MF, Mubarik M, Rejeb A, Nilashi M, Identifying industry 5.0 contributions to sustainable development: a strategy roadmap for delivering sustainability values, vol. 33. Elsevier, 2022, pp. 716–737. Accessed: Jan. 05, 2025. [Online]. https://www.sciencedirect.com/science/article/pii/S2352550922002093. Accessed 05 Jan 2025.
70. Paschek D, Luminosu CT, Ocakci E, Industry 5.0 Challenges and Perspectives for Manufacturing Systems in the Society 5.0, In: Sustainability and innovation in manufacturing enterprises: indicators, models and assessment for industry 5.0, A. Draghici and L. Ivascu, Eds., Singapore: Springer, 2022, pp. 17–63. https://doi.org/10.1007/978-981-16-7365-8_2.
71. Akundi A, Euresti D, Luna S, Ankobiah W, Lopes A, Edinbarough I, State of Industry 5.0—Analysis and identification of current research trends, vol. 5, no. 1. MDPI, 2022, p. 27. https://www.mdpi.com/2571-5577/5/1/27. Accessed 05 Jan 2025.
72. Alhaddi H. Triple bottom line and sustainability: a literature review. Bus Manage Stud. 2015;1(2):6–10.
73. Carayannis EG, Morawska-Jancelewicz J. The futures of Europe: society 5.0 and industry 5.0 as driving forces of future universities. J Knowl Econ. 2022;13(4):3445–71. https://doi.org/10.1007/s13132-021-00854-2.
74. Martín-Gómez AM, Agote-Garrido A, Lama-Ruiz JR, A framework for sustainable manufacturing: Integrating industry 4.0 technologies with industry 5.0 values, vol. 16, no. 4. Multidisciplinary Digital Publishing Institute, 2024, p. 1364. https://www.mdpi.com/2071-1050/16/4/1364. Accessed 23 Feb 2025.
75. Tallat R, et al. Navigating industry 5.0: A survey of key enabling technologies, trends, challenges, and opportunities, IEEE Communications Surveys & Tutorials, 2023, [Online]. https://ieeexplore.ieee.org/abstract/document/10305081/. Accessed 05 Jan 2025.
76. Martini B, Bellisario D, Coletti P, Human-centered and sustainable artificial intelligence in industry 5.0: Challenges and perspectives, vol. 16, no. 13. MDPI, 2024, p. 5448. [Online]. https://www.mdpi.com/2071-1050/16/13/5448. Accessed 05 Jan 2025.
77. D. S. Ambhore, "Navigating Industry 4.0 to Industry 5.0: Challenges and Strategies for Workforce Transition and its Relation to SDGs." 2024. [Online]. https://www.diva-portal.org/smash/record.jsf?pid=diva2:1837930. Accessed 23 Feb 2025.
78. Longo F, Padovano A, Umbrello S, Value-oriented and ethical technology engineering in industry 5.0: A human-centric perspective for the design of the factory of the future, vol. 10, no. 12. MDPI, 2020, p. 4182. https://www.mdpi.com/2076-3417/10/12/4182. Accessed 05 Jan 2025.
79. Dacre N, Yan J, Frei R, Al-Mhdawi MKS, Dong H. Advancing sustainable manufacturing: a systematic exploration of Industry 5.0 supply chains for sustainability, human-centricity, and resilience. Prod PlanContr. 2024. https://doi.org/10.1080/09537287.2024.2380361.
80. Abuhasel KA, A linear probabilistic resilience model for securing critical infrastructure in industry 5.0, vol. 11. IEEE, 2023, pp. 80863–80873. https://ieeexplore.ieee.org/abstract/document/10198237/. Accessed 05 Jan 2025.
81. Bhat FA, Parvez S. Emerging challenges in the sustainable manufacturing system: from industry 4.0 to industry 5.0. J Inst Eng India Ser C. 2024;105(5):1385–99. https://doi.org/10.1007/s40032-024-01046-y.
82. Laddha S, Agrawal A, Unveiling barriers to Industry 5.0 adoption in supply chains: a DEMATEL approach, vol. 59, no. 2. SciELO Brasil, 2024, pp. 123–137. [Online]. Available: https://www.scielo.br/j/rmj/a/57dJhc57rnvvVJLKnq6KZMh/. Accessed 23 Feb 2025.
83. Mukherjee AA, Raj A, Aggarwal S, Identification of barriers and their mitigation strategies for industry 5.0 implementation in emerging economies, vol. 257. Elsevier, 2023, p. 108770. https://www.sciencedirect.com/science/article/pii/S0925527323000026. Accessed 05 Jan 2025.
84. Sahoo P, Saraf PK, Uchil R. Identification of challenges to Industry 5.0 adoption in Indian manufacturing firms: an emerging economy perspective. Int J Comp Integr Manuf. 2024. https://doi.org/10.1080/0951192X.2024.2382186.
85. C.-H. Hsu, J.-C. Liu, X.-Q. Cai, T.-Y. Zhang, and W.-Y. Lv, *Enabling Sustainable Diffusion in Supply Chains Through Industry 5.0: An Impact Analysis of Key Enablers for SMEs in Emerging Economies*, vol. 12, no. 24. MDPI, 2024, p. 3938. [Online]. https://www.mdpi.com/2227-7390/12/24/3938. Accessed 23 Feb 2025.
86. Mourtzis D, Angelopoulos J, Panopoulos N, A Literature Review of the Challenges and Opportunities of the Transition from Industry 4.0 to Society 5.0, vol. 15, no. 17. MDPI, 2022, p. 6276. [Online]. https://www.mdpi.com/1996-1073/15/17/6276. Accessed 05 Jan 2025.
87. Tyagi AK, Lakshmi Priya R, Mishra AK, Balamurugan G, Industry 5.0: Potentials, Issues, Opportunities, and Challenges for Society 5.0, In: Privacy Preservation of Genomic and Medical Data, 1st ed., A. K. Tyagi, Ed., Wiley, 2023, pp. 409–432. https://doi.org/10.1002/9781394213726.ch17.
88. Wang X, Xue Y, Zhang J, Hong Y, Guo S, Zeng X, A Sustainable Supply Chain Design for Personalized Customization in Industry 5.0 Era. IEEE, 2024. [Online]. https://ieeexplore.ieee.org/abstract/document/10480134/. Accessed 05 Jan 2025.
89. Ghobakhloo M, Iranmanesh M, Morales ME, Nilashi M, Amran A. Actions and approaches for enabling Industry 5.0-driven sustainable industrial transformation: a strategy roadmap. Corp Soc Respons Env. 2023;30(3):1473–94. https://doi.org/10.1002/csr.2431.
90. Sharma R, Gupta H, Harmonizing sustainability in industry 5.0 era: Transformative strategies for cleaner production and sustainable competitive advantage, vol. 445. Elsevier, 2024, p. 141118. [Online]. https://www.sciencedirect.com/science/article/pii/S0959652624005651. Accessed 23 Feb 2025.
91. Rani S, Jining D, Shoukat K, Shoukat M, Nawaz SA, A Human–Machine Interaction Mechanism: Additive Manufacturing for Industry 5.0—Design and Management, vol. 16, no. 10. MDPI, 2024, p. 4158. [Online]. https://www.mdpi.com/2071-1050/16/10/4158. Accessed 05 Jan 2025.
92. Orlova EV, Design of personal trajectories for employees' professional development in the knowledge society under Industry 5.0, vol. 10, no. 11. MDPI, 2021, p. 427. [Online]. https://www.mdpi.com/2076-0760/10/11/427. Accessed 05 Jan 2025.

93. Gamberini L, Pluchino P. Industry 5.0: a comprehensive insight into the future of work, social sustainability, sustainable development, and career. Aust J Career Dev. 2024;33(1):5–14. https://doi.org/10.1177/10384162241231118.
94. Leng J, et al. Towards resilience in Industry 5.0: a decentralized autonomous manufacturing paradigm, vol. 71. Elsevier, 2023, pp. 95–114. [Online]. https://www.sciencedirect.com/science/article/pii/S0278612523001723. Accessed 05 Jan 2025.
95. Thoben K-D, Wiesner S, Wuest T. 'Industrie 4.0' and smart manufacturing-a review of research issues and application examples. Int J Automation Technol. 2017;11(1):4–16.
96. Antonaci FG, et al. Workplace well-being in industry 5.0: a worker-centered systematic review, vol. 24, no. 17. MDPI, 2024, p. 5473. https://www.mdpi.com/1424-8220/24/17/5473. Accessed 05 Jan 2025.
97. Kovari A, Kálmán K, Industry 5.0: Generalized definition key applications opportunities and threats, vol. 21, no. 3. 2024, pp. 267–284. http://epa.niif.hu/02400/02461/00141/pdf/EPA02461_acta_polytechnica_2024_03_267-284.pdf. Accessed 23 Feb 2025.
98. Furstenau LB, et al. Link between sustainability and industry 4.0: trends, challenges and new perspectives. Ieee Access. 2020;8:140079–96.
99. Machado CG, Winroth M, Carlsson D, Almström P, Centerholt V, Hallin M. Industry 4.0 readiness in manufacturing companies: challenges and enablers towards increased digitalization. Procedia Cirp. 2019;81:1113–8.
100. Ghobakhloo M, Iranmanesh M, Grybauskas A, Vilkas M, Petraitė M. Industry 4.0, innovation, and sustainable development: a systematic review and a roadmap to sustainable innovation. Bus Strat Env. 2021;30(8):4237–57. https://doi.org/10.1002/bse.2867.
101. Patyal VS, Sarma PRS, Modgil S, Nag T, Dennehy D. Mapping the links between Industry 4.0, circular economy and sustainability: a systematic literature review. JEIM. 2022;35(1):1–35. https://doi.org/10.1108/JEIM-05-2021-0197.
102. Thapa K, Vermeulen WJV, Deutz P, Olayide O. Ultimate producer responsibility for e-waste management–a proposal for just transition in the circular economy based on the case of used European electronic equipment exported to Nigeria. Bus Strat Dev. 2023;6(1):33–52. https://doi.org/10.1002/bsd2.222.
103. Khan M, Haleem A, Javaid M. Changes and improvements in Industry 5.0: a strategic approach to overcome the challenges of industry 4.0. Green Technol Sustain. 2023;1(2):100020.
104. Narkhede GB, Pasi BN, Rajhans N, Kulkarni A, Industry 5.0 and sustainable manufacturing: a systematic literature review. Emerald Publishing Limited, 2024. https://doi.org/10.1108/BIJ-03-2023-0196/full/html
105. Caiado RGG, Machado E, Santos RS, Thomé AMT, Scavarda LF, Sustainable I4. 0 integration and transition to I5. 0 in traditional and digital technological organisations, vol. 207. Elsevier, 2024, p. 123582. [Online]. https://www.sciencedirect.com/science/article/pii/S0040162524003780. Accessed 08 Jan 2025.
106. Sindhwani R, Behl A, Singh R, Kumari S. Can industry 5.0 develop a resilient supply chain? An integrated decision-making approach by analyzing I5.0 CSFs. Inf Syst Front. 2024. https://doi.org/10.1007/s10796-024-10486-x.
107. Kamble SS, Gunasekaran A, Sharma R. Analysis of the driving and dependence power of barriers to adopt industry 4.0 in Indian manufacturing industry. Comput Ind. 2018;101:107–19.
108. Su S, Zhong RY, Jiang Y, Song J, Fu Y, Cao H. Digital twin and its potential applications in construction industry: State-of-art review and a conceptual framework, vol. 57. Elsevier, 2023, p. 102030. https://www.sciencedirect.com/science/article/pii/S1474034623001581. Accessed 08 Jan 2025.
109. Oettmeier K, Hofmann E. Impact of additive manufacturing technology adoption on supply chain management processes and components. J Manuf Technol Manag. 2016;27(7):944–68.
110. Narkhede G, Rajhans N. An integrated approach to redesign inventory management strategies for achieving sustainable development of small and medium-sized enterprises: Insights from an empirical study in India. Bus Strat Dev. 2022;5(4):308–21. https://doi.org/10.1002/bsd2.200.